

WSN Analytics Dashboard Project Report

Automated Performance Analysis Tool for Wireless Sensor Networks

Academic Mini Project Report – 2026

1. Introduction

Wireless Sensor Networks (WSN) are widely used in smart environments, industrial monitoring, healthcare systems, agriculture, military surveillance, and IoT applications. These networks consist of multiple sensor nodes that communicate with each other to collect and transfer environmental data.

Monitoring the performance of a WSN is important because issues such as packet loss, high latency, excessive energy consumption, and reduced network lifetime can affect system reliability.

This project presents a complete WSN Analytics Dashboard developed using Python technologies. The system analyzes sensor network data, stores records in a database, generates interactive visualizations, detects network issues, and produces automated reports.

The project combines data analysis, visualization, database management, REST API integration, and report generation into a single platform.

2. Objectives

The main objectives of the project are:

- To analyze Wireless Sensor Network performance data.
 - To visualize important network metrics interactively.
 - To detect abnormal network behavior using threshold analysis.
 - To store sensor data using SQLite database.
 - To generate automated PDF reports.
 - To provide REST API support for future integrations.
 - To build a user-friendly dashboard using Streamlit.
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3. Problem Statement

Wireless Sensor Networks generate large amounts of performance-related data. Traditional methods of manually analyzing CSV files are time-consuming and inefficient.

The major challenges include:

- Difficulty in identifying weak nodes.
- High latency affecting communication.
- Energy inefficiency reducing network lifetime.
- Lack of real-time visualization tools.
- No centralized reporting system.

This project solves these problems by creating an automated analytics dashboard for sensor network monitoring.

4. Technologies Used

Technology	Purpose
Python	Core programming language
Streamlit	Dashboard development
Pandas	Data analysis and processing
NumPy	Statistical computations
Plotly	Interactive visualizations
SQLite	Database storage
SQLAlchemy	ORM database handling
Flask	REST API support
ReportLab	PDF report generation

5. System Architecture

The project workflow follows these stages:

1. CSV sensor data is uploaded.
2. Data is processed using Pandas.

3. Statistical analysis is performed.
 4. Data is stored in SQLite database.
 5. Visualizations are generated using Plotly.
 6. Dashboard displays metrics and alerts.
 7. PDF reports are generated automatically.
 8. REST API endpoints provide external access.
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6. Modules Description

6.1 Dashboard Module

The dashboard is developed using Streamlit and provides an interactive interface for users.

Features:

- Upload CSV files
 - View performance metrics
 - Filter node ranges
 - Display alerts
 - Show interactive graphs
 - Export filtered data
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6.2 Analysis Module

The analysis module performs statistical computations on sensor data.

Metrics calculated:

- Average PDR
- Average Latency
- Average Energy
- Average Lifetime
- Variance
- Standard Deviation

- Health Score

Threshold-based analysis identifies:

- LOW PDR nodes
 - HIGH LATENCY nodes
 - HIGH ENERGY nodes
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6.3 Visualization Module

Interactive visualizations are created using Plotly.

Graphs included:

- Latency Distribution Histogram
- PDR per Node Graph
- Energy Consumption Graph
- Energy vs Lifetime Bubble Chart
- Correlation Heatmap
- Radar Analysis
- Lifetime Trend Analysis

These visualizations help users easily understand network behavior.

6.4 Database Module

SQLite database is used for persistent storage.

Database fields:

Field	Description
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node_id	Sensor node identifier
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pdr	Packet Delivery Ratio
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latency	Communication delay
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energy	Energy consumption
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lifetime	Estimated operational time
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6.5 REST API Module

The Flask API allows external systems to interact with the project.

Available endpoints:

- GET /data
- POST /data
- GET /analysis
- GET /report

6.6 Report Generation Module

The ReportLab library is used to automatically generate PDF reports containing:

- Network summary
- Statistical metrics
- Alert details
- Health analysis
- Graph observations

7. Input Dataset

The dataset contains sensor network performance values.

Required columns:

- node_id
- pdr
- latency
- energy
- lifetime

The system supports CSV file uploads.

8. Key Features

- Interactive dashboard
 - Real-time filtering
 - Automated alerts
 - SQL database integration
 - REST API support
 - Advanced visualizations
 - PDF report generation
 - Health score analysis
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9. Results and Observations

The project successfully analyzed Wireless Sensor Network data and generated meaningful insights.

Observations:

- Several nodes showed LOW PDR indicating packet loss.
- Some nodes exceeded latency threshold values.
- Energy usage varied significantly between nodes.
- Weak correlation existed between energy and lifetime.
- Dashboard effectively visualized network performance.

The system successfully identified weak network areas and improved understanding of node behavior.

10. Advantages

- Easy to use interface
 - Automated analysis process
 - Efficient visualization system
 - Reduces manual effort
 - Scalable architecture
 - Suitable for IoT applications
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11. Limitations

- Uses static CSV datasets
 - No live IoT integration currently
 - SQLite suitable mainly for small-scale projects
 - No machine learning prediction model implemented
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12. Future Enhancements

Future improvements planned:

- Real-time IoT sensor integration
 - Cloud deployment
 - Machine learning-based node prediction
 - Mobile application support
 - Real-time alerts using email/SMS
 - Advanced network optimization analysis
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13. Conclusion

The WSN Analytics Dashboard project successfully demonstrates an automated system for analyzing Wireless Sensor Network performance.

The project integrates data analysis, visualization, database storage, REST API functionality, and PDF report generation into a unified platform.

The dashboard helps users identify network issues quickly and improves monitoring efficiency.

This project also demonstrates practical implementation of Python-based analytics tools for real-world IoT and sensor network applications.

14. Team Members

Name	Contribution
Janula Swetha	Dashboard & Visualization

Name	Contribution
Jinukala Vaishnavi	Data Analysis & Statistics
Pavani Reddy	Database & Backend
NJ Madhav	API & Report Generation

15. References

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